

Ayan Anees

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EDUCATION

Lahore University of Management Sciences (LUMS)

Bachelor of Science in Computer Science

Lahore, Pakistan

2023 - Expected 2027

WORK EXPERIENCE

NICL

Lahore, Pakistan

Forward Deployed Engineering Intern

June 2026 - September 2026

- Automated invoice-to-ERP handling by building a **7-stage LangGraph state machine** (OCR, extraction, validation, reconciliation, review, posting) that turns scanned documents into **Pydantic schema-validated** records and posts to the ERP only after an operator approves, replacing per-document manual entry.
- Built an HR policy assistant on **hybrid RAG** over **Databricks Vector Search**, fusing dense-vector and full-text hits by **reciprocal rank** and reranking them with a **cross-encoder**, so exact tokens such as clause numbers retrieve as reliably as paraphrased questions that pure ANN handles poorly.
- Embedded **context-enriched chunks** while quoting back only raw policy wording, making every answer **clause-cited** and forcing an explicit refusal when the documents do not cover the question, so staff never act on an invented notice period or gratuity figure.
- Made retrieval provider-agnostic behind a protocol with a local **NumPy** index fallback, letting the assistant degrade to plain ANN or run entirely offline at zero cloud spend rather than fail, and hardened ingestion with **fail-closed** admin auth, bounded retries, and per-file terminal states.

PROJECTS

- Save Syncer:** Designed a content-addressed save-file sync engine in **Python** and **FastAPI** where every snapshot is a **BLAKE2b** manifest over deduplicated blobs, so unchanged files cost nothing to re-sync and a corrupted blob is caught before it can overwrite a good copy. Replaced timestamp comparison with **monotonic revisions and three-way state**, resolving conflicts correctly even when a machine's clock runs backwards. Shipped two interchangeable remotes behind one interface: a shared folder, and a self-hosted account server with **HMAC session tokens** and **TTL locks** that stop two desktops writing the same save at once.
github.com/AyanAnees326/save-syncer
- PC Checker:** Built a Windows pre-purchase hardware inspector in **Rust** and **Tauri** that bypasses generic OS reporting to read silicon directly, **Intel/AMD MSRs** for power and thermal limits, **NVML/ADL** for GPU counters, **D3D12** for compute load, and **ATA SMART pass-through** for drive wear. Every result is an **observed-versus-expected** finding judged against the hardware's own rated figures, SMART write endurance and vendor failure thresholds, battery design capacity, and the age the seller claims, while stress runs correlate live telemetry against an **idle baseline measured on the same machine**, separating a genuine thermal defect from a harmless spike and reporting an unreadable sensor as unknown rather than as a false fault.
github.com/AyanAnees326/pc_checker
- CS Pseudo:** Engineered a browser IDE for CAIE pseudocode in **TypeScript** and **React** with a hand-written lexer, a **recursive-descent parser with error recovery** that keeps linting the rest of a broken program, and a **generator-based evaluator** that suspends mid-execution on I/O, making step-through debugging and interactive input possible entirely client-side with no server round-trip. Extended it into a teaching product with **28 guided exercises** and a second language stack, an SQL playground with its own lexer, parser, and executor, plus nearest-name hints that turn typos into actionable errors. github.com/AyanAnees326/cs-pseudo

TECHNICAL SKILLS

- AI & Retrieval:** LangGraph, hybrid RAG, reranking, Databricks Vector Search, Azure Document Intelligence
- Backend & Data:** FastAPI, REST, Server-Sent Events, Pydantic, SQLite, Delta Lake, HMAC and token auth
- Frontend & Tooling:** React, Tauri, Tailwind CSS, CodeMirror, Vite, Win32/WMI, Git, pytest, Vitest